Supplementary Tables for Vibe-Spike

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Table 1: Systematic ablation of Vibe-Spike across five EEG datasets. Results are reported as balanced accuracy (mean ± standard deviation). The **Full model** incorporates GSR reconstruction pretraining; **w/o GSR pretraining** denotes the complete architecture trained from scratch and serves as the reference for all architecture ablations. Component removal, component replacement, and simplified control experiments follow the same scratch-training protocol, with simplified controls retaining only the indicated input representation and classifier. The final column reports the average across datasets, and Δ denotes the change in percentage points relative to **w/o GSR pretraining**. Bold and underlined values indicate the best and second-best results, respectively. Statistical significance between the two was evaluated separately for each dataset using a two-sided Welch's t-test with $n = 10$ per group: $^{*}p < 0.05$, $^{**}p < 0.01$, and $^{***}p < 0.001$.

Model Variant	CHB-MIT	ISRUC	HMC	SADT	Mumtaz	Avg. (Δ)
Full model	**0.8323**** ± **0.0278**	**0.7977***** ± **0.0086**	**0.7848***** ± **0.0118**	**0.7568***** ± **0.0020**	**0.9694***** ± **0.0068**	**0.8282** (+4.03)
w/o GSR pretraining	<u>0.7816</u> ± 0.0334	0.7734 ± 0.0061	<u>0.7546</u> ± 0.0157	0.7049 ± 0.0086	<u>0.9248</u> ± 0.0121	<u>0.7879</u> (–)
Component removal and replacement under scratch training						
w/o spatial alignment	0.7783 ± 0.0391	<u>0.7740</u> ± 0.0058	0.7505 ± 0.0194	0.7027 ± 0.0108	0.9215 ± 0.0155	0.7854 (−0.25)
SA → position-based mapping	0.7345 ± 0.0362	0.7329 ± 0.0169	0.7138 ± 0.0218	0.6610 ± 0.0132	0.8686 ± 0.0177	0.7422 (−4.57)
w/o STFT tokenization	0.6814 ± 0.0362	0.2000 ± 0.0000	0.3795 ± 0.0198	0.5126 ± 0.0263	0.6008 ± 0.0187	0.4749 (−31.30)
STFT → temporal convolution	0.7629 ± 0.0349	0.7661 ± 0.0134	0.7374 ± 0.0171	0.6818 ± 0.0101	0.9052 ± 0.0135	0.7707 (−1.72)
STFT → temporal patches	0.7591 ± 0.0357	0.7697 ± 0.0132	0.7340 ± 0.0176	0.6860 ± 0.0098	0.9036 ± 0.0139	0.7705 (−1.74)
w/o Kuramoto coupling	0.7690 ± 0.0260	0.7059 ± 0.0138	0.6175 ± 0.0232	0.6857 ± 0.0140	0.8781 ± 0.0023	0.7312 (−5.66)
w/o rhythm gate	0.7704 ± 0.0341	0.7731 ± 0.0128	0.7519 ± 0.0163	<u>0.7153</u> ± 0.0091	0.9147 ± 0.0128	0.7851 (−0.28)
w/o SNN	0.7496 ± 0.0366	0.7495 ± 0.0165	0.7196 ± 0.0219	0.6649 ± 0.0124	0.8923 ± 0.0158	0.7552 (−3.27)
$\Delta t = 16$	0.7040 ± 0.0300	0.6420 ± 0.0180	0.4046 ± 0.0144	0.5395 ± 0.0193	0.6267 ± 0.0125	0.5834 (−20.45)
Simplified controls under scratch training						
STFT + classifier	0.6250 ± 0.0260	0.2380 ± 0.0120	0.2009 ± 0.0000	0.5278 ± 0.0000	0.4985 ± 0.0310	0.4180 (−36.98)
Raw + classifier	0.5740 ± 0.0240	0.2110 ± 0.0080	0.2041 ± 0.0023	0.5129 ± 0.0000	0.5352 ± 0.0081	0.4074 (−38.04)

Table 2: Effect of rhythmic gating on spike rates across five EEG datasets. Values are reported as mean ± standard deviation over ten matched seeds. Δ denotes the within-seed difference between Gate OFF and Gate ON, and relative change is calculated with respect to Gate ON. Statistical significance was evaluated separately for each dataset using a two-sided paired t-test on the within-seed differences: $^{*}p < 0.05$, $^{**}p < 0.01$, and $^{***}p < 0.001$.

Dataset	Gate ON	Gate OFF	Δ (Gate OFF − Gate ON)	Relative change (%)
HMC	0.2861 ± 0.0052	0.5299 ± 0.0083	$0.2438^{***} \pm 0.0135$	85.3350
CHB-MIT	0.3533 ± 0.0093	0.6154 ± 0.0028	$0.2621^{***} \pm 0.0120$	74.3372
ISRUC	0.2224 ± 0.0064	0.5418 ± 0.0109	$0.3193^{***} \pm 0.0173$	143.9273
Mumtaz	0.3284 ± 0.0174	0.6076 ± 0.0117	$0.2792^{***} \pm 0.0291$	85.7463
SADT	0.2995 ± 0.0080	0.6279 ± 0.0034	$0.3284^{***} \pm 0.0114$	109.8331

Table 3: Detailed statistical results supporting the performance comparisons reported in Table 2 of the main manuscript. Rank 1 and Rank 2 denote the methods with the highest and second-highest balanced accuracy on each dataset, respectively. Two-sided Welch's t-tests were reconstructed from the reported means, standard deviations, and sample sizes. Δ denotes the balanced accuracy of Rank 1 minus Rank 2 in percentage points, and N_1/N_2 denotes their respective sample sizes. No correction for multiple comparisons was applied. Significance levels are $*\,p < 0.05$, $**\ p < 0.01$, and $***\,p < 0.001$; n.s. denotes non-significance.

Dataset	Rank 1	Rank 2	N_1/N_2	Δ (pp)	t	df	p	Sig.
TUAB	REVE	CBraMod	10/10	$+0.26$	3.153	15.26	0.0065	$**$
TUEV	REVE	CBraMod	10/10	$+0.88$	1.101	12.75	0.2913	n.s.
ISRUC	Vibe-Spike	CBraMod	10/10	$+1.12$	2.537	17.01	0.0213	$*$
HMC	Vibe-Spike	REVE	10/10	$+4.47$	10.110	15.25	3.71×10^{-8}	$***$
Mumtaz2016	Vibe-Spike	REVE	10/10	$+0.50$	1.335	16.13	0.2005	n.s.
CHB-MIT	Vibe-Spike	CBraMod	10/5	$+9.25$	5.988	7.94	3.37×10^{-4}	$***$
ADFTD-RS	Vibe-Spike	REVE	10/10	$+9.14$	6.146	16.13	1.36×10^{-5}	$***$
SADT	Vibe-Spike	CNN-Transformer	10/10	$+0.07$	0.587	15.10	0.5661	n.s.
SEED-VIG	Vibe-Spike	SPaRCNet	10/10	$+0.83$	0.648	10.32	0.5312	n.s.
MentalArithmetic	REVE	Vibe-Spike	10/10	$+3.83$	2.552	17.75	0.0202	$*$

Table 4: Provenance and evaluation protocols for the performance results reported in Table 2 of the main manuscript. Published results denote values adopted from prior studies, whereas rerun results denote methods evaluated in the present study. N indicates the number of independent runs available for statistical comparison.

Benchmark setting	**Datasets**	**Published results**	**Rerun results**	**Evaluation protocol**
Protocol aligned	TUAB, TUEV, ISRUC, HMC, Mumtaz2016, MentalArithmetic	REVE Ouahidi et al. [2026] $N = 10$	Vibe-Spike $N = 10$	REVE-released preprocessed files Original benchmark splits
Partially rerun	CHB-MIT	BIOT Yang et al. [2023], $N = 5$ CBraMod Wang et al. [2025], $N = 5$	REVE, $N = 10$ Vibe-Spike, $N = 10$	BIOT-released processed files Original benchmark split
Fully controlled	ADFTD-RS, SADT, SEED-VIG	None	All methods $N = 10$	Unified preprocessing Deterministic subject-level splits Appendices D.5–D.7

Table 5: Class-specific validation of model attribution against established sleep EEG biomarkers on HMC. For correctly classified epochs in an independent confirmation subset, ρ measures the correlation between model attribution and canonical biomarker power, with 95% confidence intervals reported in brackets. Biomarker flip denotes the percentage of initially correct predictions changed by masking the specified biomarker. Random flip is averaged over three random masks with matched channel–frequency area.

Stage	**Established sleep EEG biomarker** [Troester et al., 2023]	**Attribution ρ [95% CI]**	**Biomarker flip (%)**	**Random flip (%)**
Wake	Posterior alpha activity, 8–13 Hz	0.759 [0.726, 0.789]	23.08	0.38
N1	Central theta activity, 4–8 Hz	0.461 [0.362, 0.544]	15.99	7.79
N2	Central sigma and spindle activity, 11–16 Hz	0.478 [0.441, 0.513]	80.31	11.68
N3	Frontal delta and slow-wave activity, 0.5–4 Hz	0.708 [0.675, 0.741]	14.32	0.15
REM	Central theta and sawtooth activity, 2–6 Hz	0.369 [0.311, 0.425]	98.44	2.65

Table 6: Correspondence between latent Kuramoto coherence and established EEG phase-locking biomarkers on CHB-MIT. ρ denotes the subject-level Spearman correlation between empirical pairwise phase-locking values and latent Kuramoto phase coherence in correctly classified epochs from $n = 21$ subjects. Brackets report 95% confidence intervals, and q_{BH} denotes the Benjamini–Hochberg-adjusted p-value. Only associations with confidence intervals excluding zero that remained significant after correction are reported.

Condition	**Established EEG phase-locking biomarker**	**Spearman ρ [95% CI]**	q_{BH}
Seizure beta	Interictal beta-phase synchronization [Bettus et al., 2011]	0.312 [0.240, 0.381]	0.0016
Non-seizure theta	Low-frequency phase synchronization [Bettus et al., 2008]	0.225 [0.141, 0.296]	0.0016
All-epoch alpha	Interictal alpha-phase synchronization [Quraan et al., 2013]	0.204 [0.113, 0.291]	0.0102

Table 7: Seizure-associated changes in latent Kuramoto synchrony on CHB-MIT. Statistics were computed from classified CHB-MIT epochs in each class. Significance was assessed using a two-sided label-permutation test, with the 75th-percentile phase-locking matrices and high-synchrony cores recomputed after each permutation. The observed increases are consistent with previously reported seizure-related hypersynchrony and network reorganization [Jiruska et al., 2013, Kramer et al., 2010]. $^{**}p_{\mathrm{perm}} < 0.01$; pp denotes percentage points.

Kuramoto synchrony metric	**Non-seizure**	**Seizure**	**Change**
Mean off-diagonal 75th-percentile PLV	0.8766	0.9425	$+7.52\%^{**}$
High-synchrony core density	0.9538	0.9984	$+4.46$ pp**

Table 8: Quantitative characterization of spike-module coding patterns across four EEG datasets. Event-selective and tonic-like units characterize transient and sustained spiking regimes, respectively. Effective active units are quantified by the participation ratio $(\sum_i r_i)^2/(N\sum_i r_i^2)$ over the 128 latent units. Rate modules are identified from correlations between latent-unit firing-rate profiles. Module separation is defined as $\Delta r = \bar{r}_{\text{within}} - \bar{r}_{\text{between}}$, with larger values indicating stronger within-module organization.

Dataset	Class	Event-selective units (%)	Tonic-like units (%)	Effective active units (%)	Rate modules	Module separation Δr
CHB-MIT	Non-seizure	16.41	32.81	56.92	4	0.1249
	Seizure	20.31	29.69	56.75	4	0.2406
SADT	Alert	24.22	33.59	47.82	2	0.1814
	Drowsy	21.88	32.81	54.37	4	0.2035
Mumtaz	Healthy	17.97	33.59	57.93	5	0.1414
	MDD	16.41	33.59	55.49	5	0.1156
ADFTD	CN	25.00	0.00	27.09	2	0.4270
	AD	28.12	1.56	18.58	2	0.1713

Table 9: Effect of pretraining data scale on downstream balanced accuracy across three EEG datasets. Results are reported as mean ± standard deviation. Data volume denotes the storage size of the corresponding REVE pretraining subset. The final column reports the average balanced accuracy across datasets, with Δ denoting the change in percentage points relative to scratch training. Bold values indicate the best result in each column.

Model variant	Pretraining data	Data volume	CHB-MIT	SADT	HMC	Avg. (Δ)
Scratch	None	0	0.7816 ± 0.0334	0.7049 ± 0.0086	0.7576 ± 0.0157	0.7480 (–)
Subset 1	REVE groups 100–104	9.64 GB	0.7538 ± 0.0357	0.6993 ± 0.0089	0.7526 ± 0.0190	0.7352 (−1.28)
Subset 2	REVE groups 100–110	101.42 GB	0.7927 ± 0.0290	0.7151 ± 0.0081	0.7611 ± 0.0171	0.7563 (+0.83)
Subset 3	REVE groups 100–150	301.90 GB	0.8013 ± 0.0272	0.7316 ± 0.0065	0.7614 ± 0.0185	0.7648 (+1.67)
Full	All public REVE groups	4.85 TB	**0.8323 ± 0.0278**	**0.7568 ± 0.0020**	**0.7848 ± 0.0118**	**0.7913** (+4.33)

Table 10: Class-level classification performance and error distribution of Vibe-Spike on TUEV. Support denotes the number of evaluation samples in each class. Primary error destinations indicate the two most frequent incorrect predicted classes. Macro averages are calculated without class-size weighting across the six TUEV classes.

Class	Support	Precision	Recall	F1	Specificity	Primary error destinations
SPSW	567	0.0251	0.0600	0.0354	0.9543	GPED, BCKG
GPED	4,677	0.6925	0.6665	0.6792	0.9441	BCKG, ARTF
PLED	1,998	0.4677	0.6011	0.5261	0.9502	GPED, BCKG
EYEM	329	0.5572	0.5775	0.5672	0.9948	BCKG, ARTF
ARTF	2,204	0.2869	0.6497	0.3981	0.8692	BCKG, GPED
BCKG	19,646	0.8810	0.7025	0.7817	0.8093	ARTF, PLED
Macro average	29,421	0.4851	0.5429	0.4979	–	–

Table 11: Confusion matrix of Vibe-Spike on the TUEV evaluation set. Rows represent true classes, columns represent predicted classes, and entries report the number of samples. Bold values indicate the most frequent predicted class within each row.

True class	SPSW	GPED	PLED	EYEM	ARTF	BCKG
SPSW	34	**312**	20	7	89	105
GPED	149	**3,117**	280	1	348	782
PLED	147	283	**1,201**	36	118	213
EYEM	0	0	0	**190**	42	97
ARTF	7	72	13	13	**1,432**	667
BCKG	1,017	717	1,054	94	2,962	**13,802**

Table 12: Effect of input sampling rate on downstream balanced accuracy across five EEG datasets. Results are reported as mean ± standard deviation over two paired seeds. The 100 Hz configuration serves as the reference. The final column reports the average balanced accuracy across datasets, with Δ denoting the change in percentage points relative to 100 Hz. Bold and underlined values indicate the best and second-best results for each dataset, respectively.

Sampling rate	CHB-MIT	ISRUC	HMC	SADT	Mumtaz	Avg. (Δ, pp)
100 Hz (reference)	**0.7816 ± 0.0334**	<u>0.7734</u> ± 0.0061	<u>0.7546</u> ± 0.0157	**0.7049 ± 0.0086**	**0.9248 ± 0.0121**	0.7879 (ref.)
200 Hz	<u>0.7808</u> ± 0.0335	**0.7744 ± 0.0122**	**0.7559 ± 0.0160**	<u>0.7045</u> ± 0.0086	<u>0.9244</u> ± 0.0122	0.7880 (+0.01)
50 Hz	0.7771 ± 0.0362	0.7584 ± 0.0065	0.7476 ± 0.0220	0.6964 ± 0.0088	0.9228 ± 0.0128	0.7805 (−0.74)
20 Hz	0.6971 ± 0.0384	0.7315 ± 0.0102	0.6896 ± 0.0170	0.6938 ± 0.0115	0.9008 ± 0.0150	0.7426 (−4.53)

Table 13: Representation similarity between models trained at different sampling rates on CHB-MIT, using the 100 Hz model as the reference. Results are averaged over ten paired seeds. Centered kernel alignment (CKA) is calculated for the projected time-frequency representations, Kuramoto oscillator-phase signatures, and latent-unit spike-rate representations. Higher values indicate greater representational similarity. Bold and underlined values indicate the highest and second-highest similarities for each representation, respectively.

Sampling-rate comparison	TF-projection CKA	Oscillator-phase CKA	Spike-rate CKA
100 vs. 200 Hz	**0.9792**	**0.9168**	**0.9017**
100 vs. 50 Hz	<u>0.9732</u>	<u>0.7774</u>	<u>0.7436</u>
100 vs. 20 Hz	0.9256	0.6644	0.6207

Table 14: Effect of sampling rate on latent oscillator dynamics on CHB-MIT. Results are reported as mean ± standard deviation over ten paired seeds. Temporal pairwise PLV denotes the phase-locking value computed over the sampled oscillator-unit subset, and R_1 denotes the first-order Kuramoto order parameter.

Sampling rate	Temporal pairwise PLV	Kuramoto order R_1
200 Hz	0.9297 ± 0.0151	0.8304 ± 0.0126
100 Hz	0.9364 ± 0.0245	0.8527 ± 0.0070
50 Hz	0.8537 ± 0.0051	0.8047 ± 0.0072
20 Hz	0.8376 ± 0.0118	0.7763 ± 0.0035

Table 15: Input size and model-side computational cost at 100 and 200 Hz under matched experimental configurations. All measurements use 30 s EEG epochs. The final column reports the ratio between the 200 Hz and 100 Hz measurements.

Quantity	**100 Hz**	**200 Hz**	**Ratio (200/100)**
Raw samples per 30 s epoch	3,000	6,000	2.00
Raw input per batch (MB)	0.87	1.74	2.00
Nyquist frequency (Hz)	50	100	2.00
STFT frequency bins	65	129	1.99
STFT temporal frames	188	188	1.00
Inference latency (ms/sample)	24.93	24.94	1.00
Inference throughput (samples/s)	40.11	40.10	1.00
Inference activation peak (MB)	27.88	49.68	1.78
Training-step time (ms/batch)	451.20	464.74	1.03
Training activation peak (MB)	478.45	502.51	1.05

Table 16: Post-hoc CPU and GPU inference latency of Vibe-Spike and four EEG foundation-model baselines on 60 s HMC segments. All measurements used a batch size of 1, 100 warm-up iterations, 100 timed iterations, and ten independent rounds. CPU inference was evaluated on an Intel Xeon Gold 6459C using one PyTorch thread, whereas GPU inference was evaluated on a single NVIDIA GeForce RTX 5090. Median and P95 latency are reported. The real-time factor is calculated as $\mathrm{RTF} = t_{\mathrm{inference}}/T_{\mathrm{EEG}}$, where $T_{\mathrm{EEG}} = 60\,\mathrm{s}$. Bold values indicate the lowest median latency, P95 latency, and RTF on each device.

Model	**EEG duration (s)**	**CPU: Intel Xeon Gold 6459C**		**GPU: NVIDIA GeForce RTX 5090**		**RTF**	
		Median (ms)	**P95 (ms)**	**Median (ms)**	**P95 (ms)**	**CPU**	**GPU**
Vibe-Spike	60	168.850	172.542	181.624	184.305	0.002814	0.003027
REVE	60	619.910	637.244	6.842	6.910	0.010332	0.000114
BIOT	60	**23.224**	**24.634**	**1.248**	**1.280**	**0.000387**	**0.000021**
LaBraM	60	33.760	34.113	9.028	9.495	0.000563	0.000150
CBraMod	60	41.018	42.424	6.695	6.786	0.000684	0.000112

Table 17: Runtime latency of Vibe-Spike and four EEG foundation-model baselines on a Raspberry Pi 5 (16 GB). Post-hoc measurements use full inference on 60 s HMC segments, whereas real-time monitoring uses a 2 s update interval. Median and P95 latency are reported per operation. Bold values indicate the lowest latency and RTF within each application setting.

Model	**Application setting**	**Operation**	**Update interval (s)**	**Median (ms/op)**	**P95 (ms/op)**	**Miss rate (%)**	**RTF**
Vibe-Spike	Post-hoc analysis	Full inference	–	730.214	764.233	0.0	0.0122
REVE	Post-hoc analysis	Full inference	–	2681.913	2724.687	0.0	0.0447
BIOT	Post-hoc analysis	Full inference	–	**107.909**	**138.208**	0.0	**0.0018**
LaBraM	Post-hoc analysis	Full inference	–	158.847	190.737	0.0	0.0026
CBraMod	Post-hoc analysis	Full inference	–	199.580	233.524	0.0	0.0033
Vibe-Spike	Real-time monitoring	Stream update	2	**29.779**	**44.941**	0.0	**0.0149**
REVE	Real-time monitoring	Stream update	2	2872.689	3090.000	100.0	1.4363
BIOT	Real-time monitoring	Stream update	2	107.174	137.630	0.0	0.0536
LaBraM	Real-time monitoring	Stream update	2	163.333	194.150	0.0	0.0817
CBraMod	Real-time monitoring	Stream update	2	208.256	242.210	0.0	0.1041

Table 18: Whole-system power and inference energy of Vibe-Spike and four EEG foundation-model baselines on a Raspberry Pi 5 (16 GB). Measurements were obtained using an external power meter as the board's sole power source and averaged over three 15 min runs. Experiments used HMC samples with a batch size of 1 and covered post-hoc full inference and real-time stream updates scheduled every 2 s. Run energy was calculated as $E_{\mathrm{run}} = (Wh_{\mathrm{end}} - Wh_{\mathrm{start}}) \times 3600$. The measured idle power of 2.94 W was subtracted to obtain power and energy above idle. Bold values indicate the lowest system power, power above idle, and energy above idle within each application setting.

Model	Application setting	Operation	Idle power (W)	System power (W)	Power above idle (W)	Energy above idle (J/op)
Vibe-Spike	Post-hoc analysis	Full inference	2.940	**3.558**	**0.618**	**1.236**
REVE	Post-hoc analysis	Full inference	2.940	7.392	4.452	17.808
BIOT	Post-hoc analysis	Full inference	2.940	4.260	1.320	2.640
LaBraM	Post-hoc analysis	Full inference	2.940	5.862	2.922	5.844
CBraMod	Post-hoc analysis	Full inference	2.940	6.084	3.144	6.288
Vibe-Spike	Real-time monitoring	Stream update	2.940	**3.420**	**0.480**	**0.960**
REVE	Real-time monitoring	Stream update	2.940	6.431	3.491	10.022
BIOT	Real-time monitoring	Stream update	2.940	4.350	1.410	2.820
LaBraM	Real-time monitoring	Stream update	2.940	6.312	3.372	6.744
CBraMod	Real-time monitoring	Stream update	2.940	4.170	1.230	2.460

Table 19: Estimated per-operation computational cost of Vibe-Spike and four EEG foundation-model baselines using the same batch-1, 60 s HMC input as the CPU and GPU latency experiments. FLOPs are obtained using the PyTorch operator profiler, and MAC-equivalent operations are calculated assuming two FLOPs per multiply-accumulate. Arithmetic-energy estimates use 4.6 pJ/MAC and 0.9 pJ/SynOp for a normalized 45-nm FP32 implementation [Cao et al., 2024, Horowitz, 2014]. Readout SynOps are reported separately from MAC-equivalent operations and apply only to Vibe-Spike. Bold values indicate the lowest FLOPs, MAC-equivalent operations, and MAC-equivalent energy. These estimates represent arithmetic-cost proxies rather than measured system energy.

Model	FLOPs/op (G)	MAC-eq./op (G)	Readout SynOps/op (K)	MAC-eq. energy/op (mJ)	Readout SynOp energy/op (μJ)
Vibe-Spike	**0.103**	**0.051**	12.528	**0.236**	0.0113
REVE	75.838	37.919	–	174.428	–
BIOT	3.085	1.542	–	7.095	–
LaBraM	2.772	1.386	–	6.376	–
CBraMod	3.808	1.904	–	8.759	–

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